Circles and Parabolas in Standard Form
Answer Key

Algebra 2

Quick Answers

1. 25, $(x-5)^2$
2. $(x-3)^2+(y+2)^2-25$, 5
3. $(x+4)^2+(y-1)^2-9$, 3
4. $(x-4)^2-5$, -5
5. $(x+3)^2-4$, -5, -1
6. $2\,(x-2)^2+2\,(y+3)^2-50$, 5
7. $2\,(x-3)^2-5$, 3
8. $(x+5)^2+(y-3)^2-16$, 3.61, —
9. x^2+2x+y^2-8y-3
10. $(y-2)^2+3$, 2
11. $x=3$, $y=4$, $x=$ -4, $y=$ -3
12. $(x-2)^2+(y+3)^2+7$

Curriculum

Level: Algebra 2
HSA-REI.C.7 – *problem 11*
HSF-IF.C – *problems 4, 5, 7, 10*
HSG-GPE.A.1 – *problems 1, 2, 3, 6, 8, 9, 12*
Difficulty 1–5 of 5 across 23 tagged checks.

Common wrong answers

2. *If they got 25: stopped at the number on the right-hand side and reported r squared as the radius*
7. *If they got 6: completed the square without first factoring the 2 out of both x terms, halving 12 instead of 6*
10. *If they got 3: wrote the vertex as (2, 3) by giving the coordinates in the usual order instead of noticing that the completed square is in y*

What this sheet is teaching, and what to listen for. One move — completing the square — does all twelve problems, and the sheet varies what surrounds it: two variables instead of one, a coefficient in front of the squared term, a square completed in y rather than x, and finally a right-hand side that turns out negative. When a student's answer is wrong, ask which number they halved and what they did to the other side. Almost every error here is a balance error or a sign read backwards off $(x+h)^2$.

1. The completing move on its own. $x^2-10x+\_\_=(x-\_)^2$

Half of -10 is -5, and $(-5)^2=25$, so the number to add is $\boxed{25}$

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That makes $x^2-10x+25=\boxed{(x-5)^2}$.

The pattern to carry through the sheet. x^2+bx needs $\left(\frac{b}{2}\right)^2$ added, and the result is always $\left(x+\frac{b}{2}\right)^2$ — the number inside the bracket is half of b, sign and all, never b itself.

2. A circle from general form. $x^2 + y^2 - 6x + 4y - 12 = 0$

Group and move the constant: $(x^2 - 6x) + (y^2 + 4y) = 12$.
Half of -6 is -3, so add 9; half of 4 is 2, so add 4. Both must go on the right too:

$$(x^2 - 6x + 9) + (y^2 + 4y + 4) = 12 + 9 + 4$$

$$\boxed{(x-3)^2 + (y+2)^2 = 25}$$

Centre $(3, -2)$, and the radius is $\sqrt{25} = \boxed{5}$.
Read the centre with the signs flipped. $(y+2)^2$ is $(y-(-2))^2$, so $k = -2$. A student who reports $(3, 2)$ has read the printed sign rather than the form.
If they answered 25 for the radius: the number on the right of a circle equation is r^2. Ask them what the radius of $x^2 + y^2 = 9$ is; almost everyone says 3, and then the general rule lands.

3. Watch the signs. $x^2 + y^2 + 8x - 2y + 8 = 0$

Group and move the constant: $(x^2 + 8x) + (y^2 - 2y) = -8$.
Half of 8 is 4, so add 16; half of -2 is -1, so add 1:

$$(x^2 + 8x + 16) + (y^2 - 2y + 1) = -8 + 16 + 1$$

$$\boxed{(x+4)^2 + (y-1)^2 = 9}$$

Centre $(-4, 1)$, radius $\sqrt{9} = \boxed{3}$.
A negative constant is not a problem. Here the right-hand side starts at -8 and the two added squares carry it up to 9. Students often assume they have made a mistake when the right side is briefly negative; problem 12 is the case where it stays negative, which is a different matter entirely.

4. A parabola into vertex form. $y = x^2 - 8x + 11$

Half of -8 is -4, and $(-4)^2 = 16$. Add and subtract it in the same expression so nothing changes:

$$y = (x^2 - 8x + 16) - 16 + 11$$

$$\boxed{y = (x-4)^2 - 5}$$

Vertex $(4, \boxed{-5})$.
This balances differently from a circle. A circle equation has two sides, so the added 16 goes on the right as well. Here there is only one expression, so the 16 has to be subtracted again immediately. Mixing the two habits is the commonest error on this sheet — watch for a student who adds 16 and never takes it away.

5. Vertex form and the x-intercepts. $y = x^2 + 6x + 5$

Half of 6 is 3, and $3^2 = 9$:

$$y = (x^2 + 6x + 9) - 9 + 5 \qquad \Longrightarrow \qquad \boxed{y = (x+3)^2 - 4}$$

Vertex $(-3, -4)$.
For the intercepts set $y = 0$: $(x+3)^2 = 4$, so $x + 3 = \pm 2$, giving $x = \boxed{-1}$ and $x = \boxed{-5}$.
Two routes, one answer. Factoring $x^2 + 6x + 5 = (x+5)(x+1)$ gives the same two intercepts in one line. Vertex form earns its keep when the quadratic does not factor — and it also hands you the vertex, which factoring does not.
A check worth doing. The vertex sits halfway between the intercepts: halfway between -5 and -1 is -3, which matches $h = -3$. That symmetry is free and catches sign slips.

6. A circle with a coefficient in front. $2x^2 + 2y^2 - 8x + 12y - 24 = 0$

Divide everything by 2 first — the standard form has no coefficient in front of the squares:

$$x^2 + y^2 - 4x + 6y - 12 = 0$$

Then group and complete: half of -4 is -2 (add 4), half of 6 is 3 (add 9):

$$(x^2 - 4x + 4) + (y^2 + 6y + 9) = 12 + 4 + 9$$

$$\boxed{(x-2)^2 + (y+3)^2 = 25}$$

Centre $(2, -3)$, radius $\sqrt{25} = \boxed{5}$.
Divide first, always. Completing the square inside $2x^2 - 8x$ without factoring the 2 out leads to a right-hand side that is twice what it should be, and the radius comes out $\sqrt{2}$ times too big. Dividing the whole equation by 2 at the start makes the coefficient disappear for good; there is nothing to remember later.

7. A parabola with a coefficient in front. $y = 2x^2 - 12x + 13$

Here the 2 cannot simply be divided away — it belongs to the function. Factor it out of the x terms only:

$$y = 2(x^2 - 6x) + 13$$

Half of -6 is -3, so add 9 inside the bracket. Because the bracket is multiplied by 2, that adds 18 overall, and 18 must come back off:

$$y = 2(x^2 - 6x + 9) - 18 + 13 \qquad \Longrightarrow \qquad \boxed{y = 2(x-3)^2 - 5}$$

Vertex $(\boxed{3}, -5)$.
If they answered 6 for the x-coordinate: they halved -12 instead of -6, which means they completed the square before factoring the 2 out. Once the bracket reads $x^2 - 6x$, the number to halve is -6.
The -18 is where marks are lost. Adding 9 inside a bracket multiplied by 2 does not add 9 to the function, it adds 18. Expanding the final answer back out is a five-second check that catches this every time.

8. Inside, on, or outside? $x^2 + y^2 + 10x - 6y + 18 = 0$, and the point $(-2, 1)$.
Half of 10 is 5 (add 25); half of -6 is -3 (add 9):

$$(x^2 + 10x + 25) + (y^2 - 6y + 9) = -18 + 25 + 9$$

$$\boxed{(x+5)^2 + (y-3)^2 = 16}$$

Centre $(-5, 3)$, radius 4.
Distance from the centre to $(-2, 1)$:

$$\sqrt{(-2-(-5))^2 + (1-3)^2} = \sqrt{9+4} = \sqrt{13}$$

$\boxed{\sqrt{13} \approx 3.61 < 4}$, so the point is nearer the centre than the radius: it lies **inside** the circle.
Standard form is the whole point of the question. From the general form there is no way to see where the centre is, so the distance cannot be computed at all. Rewriting is not a presentational step here — it is what makes the question answerable.

9. Back the other way. $(x+1)^2 + (y-4)^2 = 20$
Expand both squares and collect:

$$x^2 + 2x + 1 + y^2 - 8y + 16 = 20$$

$$\boxed{x^2 + y^2 + 2x - 8y - 3 = 0}$$

Why this direction is worth practising. It is the check on every other problem on the sheet: expanding a standard form must give back the general form you started from. A student who runs this check on problems 2, 3 and 6 never has to wonder whether they balanced correctly.

10. A parabola on its side. $x = y^2 - 4y + 7$
The square is completed in y, because that is the variable being squared. Half of -4 is -2, and $(-2)^2 = 4$:

$$x = (y^2 - 4y + 4) - 4 + 7 \qquad \Longrightarrow \qquad \boxed{x = (y-2)^2 + 3}$$

The form says x is smallest when $y = 2$, and then $x = 3$. Written as an ordered pair the vertex is $\boxed{(3,\ 2)}$, and the parabola opens to the right.
If they answered $(2, 3)$**:** they read the numbers off in the order they appear in the form, as they would for $y = (x-h)^2 + k$. Here the completed square is in y, so the number inside the bracket is the y-coordinate. Have them substitute $y = 2$ and check that x really does come out as 3 — a point that fails substitution is not on the curve.

11. Where the line meets the circle. $x^2 + y^2 = 25$ and $y = x + 1$

Substitute the line into the circle:

$$x^2 + (x+1)^2 = 25 \quad \Longrightarrow \quad 2x^2 + 2x - 24 = 0 \quad \Longrightarrow \quad x^2 + x - 12 = 0$$

Factor: $(x-3)(x+4) = 0$, so $x = 3$ or $x = -4$.
Back into $y = x + 1$: when $x = 3$, $y = 4$; when $x = -4$, $y = -3$.
The intersection points are $\boxed{(3,\,4)}$ and $\boxed{(-4,\,-3)}$.
Substitute both values back before writing them down. $3^2 + 4^2 = 25$ and $(-4)^2 + (-3)^2 = 25$, so both points really are on the circle. Solving only for x and stopping is the commonest omission here — a point needs two coordinates.

12. A circle that isn't there. $x^2 + y^2 - 4x + 6y + 20 = 0$

(a) Half of -4 is -2 (add 4); half of 6 is 3 (add 9):

$$(x^2 - 4x + 4) + (y^2 + 6y + 9) = -20 + 4 + 9$$

$$\boxed{(x-2)^2 + (y+3)^2 = -7}$$

(b) (Open explanation — listen to the reason.)
A model answer. "There is no graph. The left-hand side is a sum of two squares of real numbers, and a square is never negative, so the smallest value the left side can take is 0 — it can never equal -7. That means no point (x, y) in the plane satisfies the equation, so the equation has an empty graph. It looks like a circle equation in general form, but completing the square shows that r^2 would have to be negative, and no circle has an imaginary radius."
Marking guide. Full credit for saying the graph is empty (no real points, no circle) *and* giving the reason — a sum of squares cannot be negative, or equivalently $r^2 < 0$ is impossible. Partial credit for noticing that -7 is impossible for r^2 without saying what that means for the graph. Not sufficient: "the radius is $\sqrt{-7}$" offered as the whole answer, and definitely not "it is a circle of radius -7".
Why this problem is here. Every other problem on the sheet ends with a picture, which trains students to believe that any equation of this shape is a circle. It is not: completing the square is what decides the question, and the three outcomes are a circle ($r^2 > 0$), a single point ($r^2 = 0$), and nothing at all ($r^2 < 0$). Asking for the $r^2 = 0$ case is a good follow-up — $x^2 + y^2 - 4x + 6y + 13 = 0$ is the same problem with the one point $(2, -3)$ as its entire graph.